Geometric Annihilation of the $GM_\odot$ Parameter: Kinematic Origin of the 31^6 Exponent

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Abstract

We present a mass-free, kinematic deconstruction of the classical solar gravitational parameter $GM_\odot$ utilizing the Universal Kochenov Framework (UKF). We demonstrate that the composite Newtonian dimension $[\mathrm{m}^3/\mathrm{s}^2]$ is a methodological artifact arising from the use of an anthropogenic time scale to measure a discrete spatial lattice. By redefining the classical second as a physical projection of the Earth's equatorial waveguide step ($u^* \approx 463.83$ m), the gravitational parameter is shown to collapse into a pure topological length. Within this register-generated architecture, the macro-scale exponent 31^6 in the system boundary address ($\ell_{31} = 31^6 \cdot r_{\mathrm{AU}}$) is shown to be non-arbitrary. It emerges rigorously as the geometric square of the cubic kinematic velocity core ($31^3 \approx v_\oplus$) governed by the classical vis-viva relation. Furthermore, we show that the third terrestrial node acts as a volumetric closure of the inner planetary system, satisfying a rigid dimensionless potential identity driven by the operator $2D(3) = 2(3 - \phi^{-1})$. The numerical alignment, bounded by the elasticity residual $\delta_{\mathrm{num}} \approx 0.0424\%$, between the global register boundary and the local solar anchor reveals the precise scale shift masked by the classical meter. Finally, the prime number 31 is established as a scale-invariant structural chord of the lattice, bridging laboratory-scale torsion-balance determinations ($G_{31} = 31^2/144$), intermediate orbital velocity (31^3), and the global solar macro-address (31^6).

1 Historical Dimensional Error: The Artificial Second

In classical Newtonian mechanics, the motion of celestial bodies is described by the dynamical equation:

$$GM_\odot = 4\pi^2 \frac{r^3}{T^2}, \tag{1}$$

which attributes an artificial composite dimension of $[\mathrm{m}^3/\mathrm{s}^2]$ to the gravitational parameter. This formulation necessitates the introduction of an abstract entity—"mass" (M)—and an independent arrow of time (t).

However, the physical reality of the gauge anchor (Earth) demonstrates that time is not an independent coordinate, but rather represents a metric step of the spatial lattice. The classical second is merely a projection of the spatial waveguide step onto an anthropogenic time scale.

2 Geometric Equivalent of the Second

The spatial nature of the second is determined through the kinematic parameters of the Earth's equatorial contour. The equatorial perimeter (the boundary of the local waveguide) is $L_{\mathrm{Earth}} \approx 40,075,017$ m. In dimensionless kinematics, time is quantized by the physical rotation of the

planet relative to the carrier frequencies of the field. This generates the solar spatial step:

$$u^* = \frac{40,075,017 \text{ m}}{86,400} \approx 463.83 \text{ m}. \tag{2}$$

Here, the denominator $86,400 = 120 \times 6!$ is a dimensionless clock cycle counter, and the historical SI second is replaced by the physical segment of the spatial lattice. Time in this framework is a metric length.

3 Kinematic Origin of the 31^6 Exponent

The central question is why the register-generated scale involves the sixth power of the Mersenne number $M_5 = 31$. The answer is algebraic rather than mystical, arising directly from the vis-viva relation for a circular orbit:

$$GM_\odot = v_\oplus^2 \cdot r_{\text{AU}}. \tag{3}$$

Within the UKF architecture, the orbital velocity of the Earth is not a dynamical consequence of mass attraction by the Sun, but a kinematic projection of the cubic register core:

$$v_\oplus \approx 31^3 = 29,791 \text{ m/s}. \tag{4}$$

The observed kinematic velocity of the Earth ($v_\oplus^{\text{obs}} \approx 29,784.7$ m/s) deviates from this core by merely $\delta_v \approx 0.021\%$.

Substituting the cubic core into the vis-viva relation directly yields the topological macro-address of the system boundary:

$$GM_\odot \approx (31^3)^2 \cdot r_{\text{AU}} = 31^6 \cdot r_{\text{AU}}. \tag{5}$$

The exponent 6 is not an arbitrary power; it is the geometric square of the kinematic velocity core. The orbital velocity of a planet is the square root of the topological address of space.

4 Numerical Correspondence in SI Units

We calculate the register-generated spatial scale:

$$\ell_{31} = 31^6 \cdot r_{\text{AU}} = 887,503,681 \cdot 1.495978707 \times 10^{11} \text{ m} \approx 1.32768 \times 10^{20} \text{ m}. \tag{6}$$

To compare numerical values without equating quantities of different dimensions, we define the dimensionless ratios $L_{31} = \ell_{31}/(1 \text{ m})$ and $G_\odot = GM_\odot/(1 \text{ m}^3\text{s}^{-2})$. For the accepted baseline value of $G_\odot = 1.327124 \times 10^{20}$, their relative numerical discrepancy is:

$$\delta_{\text{num}} = 100 \left(\frac{L_{31}}{G_\odot} - 1 \right) \% \approx 0.0424\%. \tag{7}$$

The identity $GM_\odot \approx 31^6 \cdot r_{\text{AU}}$ is not coincidental, as both constants are derivatives of the geometry of the exact same gauge waveguide—the Earth. The $GM_\odot$ parameter in classical mechanics is not a fundamental mass constant, but a topological length $31^6 \cdot r_{\text{AU}}$ masked as a composite dynamic dimension through the square of the metric step $(u^*)^2$.

5 Level $n = 3$: Volumetric Closure and Dimensionless Calibration Anchor

Within the architecture of the dimensionless UKF register, coordinate units and anthropogenic artifacts such as the classical meter completely annihilate. The macrostructure of the Solar System manifests as a self-generating fractal network, where the orbital scale of the Earth is not an isolated dynamical coincidence, but a rigorous algebraic closure of the inner quantum shell.

5.1 The Terrestrial Algebraic Closure

The boundary between the two innermost planetary nodes represents a rigid algebraic identity. Upon subtracting the inverse-square address fields of Mercury and Venus, their potential difference structurally dictates the generation of the third node (Earth):

$$\Delta A = \left(\frac{1}{a_{\text{Mer}}}\right)^2 - \left(\frac{1}{a_{\text{V}}^{(\phi)}}\right)^2 \approx 2D(3) = 2(3 - \phi^{-1}), \tag{8}$$

where $a_V^{(\phi)} = 1/D(2) = 1/(3-\phi) \equiv \phi/\sqrt{5} \approx 0.723607$ AU denotes the strictly derived Venus address (observed value 0.723332 AU; deviation 0.038%), and $D(3) = 3 - \phi^{-1}$ is the volumetric operator of the third node.

Evaluating pure register values yields:

$$6.673554 - 1.909830 = 4.763724 \longleftrightarrow 2 \times 2.381966 = 4.763932. \tag{9}$$

The relative residual is strictly bounded at $\delta \approx -0.00437\%$. Within the UKF register, the level difference is retained internally as a closed economy: it condenses into the next address node (Mercury $-$ Venus $\longrightarrow$ Earth). Earth acts as the definitional metric lock ($a_\oplus \equiv 1$ AU), at which the register shifts from a planar to a volumetric (3D) state.

5.2 Absolute Scaling of the Solar Address

To project this topological architecture onto the classical gravitational parameter $GM_\odot$ without introducing mass or independent time (t), we define the absolute dimensionless scale of the Sun ($N_\odot$) strictly in units of the fundamental spatial waveguide step $u^* \approx 463.83$ m (the equatorial quantum of the terrestrial anchor).

The artificial SI dimension [m^3/s^2] is fully eliminated by dividing the empirical parameter $GM_\odot$ by the cubic power of the spatial lattice step $(u^*)^3$:

$$N_\odot = \frac{GM_\odot}{(u^*)^3} = \frac{1.327124 \times 10^{20}\ \text{m}^3\text{s}^{-2}}{(463.831215\ \text{m})^3} \approx 1.329938 \times 10^{12}. \tag{10}$$

The resulting value $N_\odot$ is a pure, dimensionless topological count. In the register convention, the classical second is a dimensionless tick count (86,400 ticks per mean solar day) and u^* carries the dimension of metre; $N_\odot$ is therefore a pure topological count.

According to the register geometry, the macroscopic system boundary is governed by the 31^6 exponent. Let R_{AU} be the normalized address of the astronomical unit expressed in the same quantum steps of the waveguide:

$$R_{\text{AU}} = \frac{r_{\text{AU}}}{u^*} = \frac{1.495978707 \times 10^{11}\ \text{m}}{463.831215\ \text{m}} \approx 322,526,569.5. \tag{11}$$

The total register-generated volume at the boundary is given by:

$$N_{\text{reg}} = 31^6 \cdot R_{\text{AU}} = 887,503,681 \cdot 322,526,569.5 \approx 2.86243 \times 10^{17}. \tag{12}$$

The direct scaling ratio between the global register boundary (N_{reg}) and the local volumetric solar anchor ($N_\odot$) reveals the exact spatial distortion factor masked by classical mechanics:

$$\frac{N_{\text{reg}}}{N_\odot} = \frac{2.86243 \times 10^{17}}{1.329938 \times 10^{12}} \approx 215,230. \tag{13}$$

Notice that $N_{\text{reg}}/N_\odot \approx 215,230 \approx (u^*/1\ \text{m})^2 \cdot (L_{31}/G_\odot)$, i.e. the square of the metric step multiplied by the same elasticity residual $\delta_{\text{num}} \approx 0.0424\%$ of Eq. (7). The recurrence of the identical residual closes the consistency circuit between the local anchor and the global boundary.

This identity demonstrates that the anthropogenic "meter" was implicitly locked inside the classical equations as an uncompensated scale shift between the local terrestrial waveguide and the global 31^6 register. Once coordinate units completely cancel out, the parameter $GM_\odot$ is stripped of its dynamical interpretation and returns to a pure combinatoric address within a self-generating spatial lattice. The fully dimensionally consistent translation between SI and register units is deferred to the Companion Note (Metric Bridge).

5.3 The Double Lock: Laboratory Boundary and Mercurial Address

The foundational boundary condition of the terrestrial register is provided by the inverse-square address field of Mercury, a_{Mer}^{-2}. This address maps precisely onto the fundamental integer ratio of the primary register:

$$G_{31} = \frac{31^2}{144} = \frac{961}{144} = 6.6736111\ldots \tag{14}$$

where $144 = F_{12} = 12^2$ is the twelfth Fibonacci node, anchoring the combinatorial lattice. This pure, dimensionless register ratio governs the exact numerical range of Newton's gravitational constant G when scaled by the classical factor of 10^{-11}. The double lock manifests when comparing the raw Mercurial address to the register ratio:

$$\frac{1}{a_{\mathrm{Mer}}^2} = \frac{1}{(0.38709843)^2} = 6.6735540\ldots \tag{15}$$

The relative deviation between the observed orbital parameter and the pure register chord is strictly bounded at $\delta \approx -0.000855\%$. Equation (15) is an algebraic identity of the register chords; its physical content lies in the bounded deviation from the observed planetary metric.

5.4 The 12/31 Reciprocal Invariant: Absolute Geometric Inversion of the Mercurial Node

The combinatorial calibration of the local laboratory register constant $G_{31} = 31^2/144$ (Eq. 14) reveals that the denominator is anchored directly to the twelfth Fibonacci node ($12^2 = 144$), which structurally conditions the underlying spatial lattice. However, the absolute geometric closure of the inner planetary architecture manifests when evaluating the bare coordinate relationship between this gravitational projection and the primary boundary condition of the system—the semi-major axis of the first planetary node (Mercury).

In classical dynamical astrophysics, the orbital address of Mercury ($a_{\mathrm{Mer}}^{\mathrm{obs}} \approx 0.387098$ AU) is interpreted as a stochastic, chaotic artifact of planetesimal accretion disk mechanics. Within the coordinate constraints of the Universal Kochenov Framework (UKF), this empirical free parameter completely annihilates. It emerges rigorously as a pure, dimensionless ratio of the fundamental structural chords of the lattice:

$$a_{\mathrm{Mer}} \equiv \frac{12}{31} \approx 0.38709677 \text{ AU}. \tag{16}$$

The relative numerical deviation between this discrete register chord and the observed astronomical baseline is tightly bounded by the exceptional constraint:

$$\delta_{\mathrm{orbit}} = 100 \left(\frac{a_{\mathrm{Mer}}^{\mathrm{obs}}}{a_{\mathrm{Mer}}} - 1 \right) \% \approx 0.0003\%. \tag{17}$$

This exact definition of the first planetary address exposes a profound, structurally sealed reciprocal inversion that governs the framework's local metric scaling. By direct comparison of the Mercurial spatial node (a_{Mer}) with the quadratic register calibration factor G_{31}, we establish a rigid, mutually inverse algebraic identity:

$$G_{31} = \frac{31^2}{144} = \left(\frac{31}{12} \right)^2 \equiv \frac{1}{a_{\mathrm{Mer}}^2}. \tag{18}$$

The first planetary node is anchored directly onto the bare integer fraction 12/31 relative to the terrestrial metric lock ($a_\oplus \equiv 1$ AU), while the laboratory manifestation of cosmic tension (G) is determined by the quadratic inversion of that identical register coordinate.

5.5 Fractal Scale-Invariance and the Hierarchical Chord

The appearance of the 31^6 exponent at the global system boundary is neither an isolated empirical coincidence nor a localized fitting parameter. Within the UKF register architecture, the prime number 31 acts as a scale-invariant topological chord that bridges laboratory-scale metrics, inner planetary nodes, and global boundary parameters.

Consequently, the whole terrestrial register exhibits an uncompromised algebraic hierarchy based entirely on progressive powers of the 31 operator:

- **The Foundational Address Chord:** The rational planetary address of the first node: $a_{\text{Mer}} = 12/31$.
- **The Micro/Laboratory Boundary (G_{31}):** The quadratic inversion of the spatial address: $(31/12)^2 = 31^2/144$.
- **The Intermediate Kinematic Velocity ($v_\oplus$):** The cubic kinetic projection of the core operator: 31^3.
- **The Macro System Boundary ($GM_\odot$):** The volumetric closure of the boundary register: $(31^3)^2 = 31^6$.

By establishing the integer invariants $\{12, 31\}$ as the prime generative constraints of the lattice, the UKF eliminates the requirement for historical dynamic adjustments. The mathematical reciprocity demonstrated herein confirms that the geometry of the Universe ignores the abstract variables of mass; it strictly obeys the combinatoric address of the lattice.

6 Conclusion

The spatial geometry of the Solar System is governed by the address code of a combinatoric register. The parameter $GM_\odot$ is deconstructed and returned to pure geometry as a macroscopic linear address of the system boundary: $31^6 \cdot r_{\text{AU}}$. The exponent 6 arises algebraically as the square of the cubic kinematic velocity core $31^3 \approx v_\oplus$, linked via the vis-viva relation. By eliminating the artificial second and the concept of mass, UKF shows that the dynamical reading of "gravitational attraction" can be replaced, without loss of numerical content, by the topological tension of the spatial lattice.

The metric ignores mass; it obeys the address.

Status: Original Architecture (Zero Fitted Parameters)

In the $GM_\odot$ correspondence no constant is fitted: the inputs are measured values (r_{AU}, equatorial perimeter, day length, torsion-balance determinations of G) and the register integers $\{12, 31, 144, 720, \phi\}$; the residuals $\delta_{\text{num}} \approx 0.0424\%$ and $\delta \approx -0.00437\%$ are outputs of the correspondence, not inputs.

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